

Chapter 10: Vector Algebra & Spatial Geometry

Advanced Mathematical Study • Vectors, Coordinate Systems & Spatial Calculus

10.1 INTRODUCTION TO VECTORS IN THREE-DIMENSIONAL SPACE ($\mathbb{R}^3$)

A vector is a mathematical object possessing both magnitude and direction. In three-dimensional Cartesian coordinate space ($\mathbb{R}^3$), a vector v is represented relative to three mutually perpendicular axes (x , y , and z) governed by standard right-hand orientation.

Definition: Component Form & Position Vectors

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two points in $\mathbb{R}^3$. The directed segment from P to Q forms vector $v = \langle v_1, v_2, v_3 \rangle$ where:

$$v_1 = x_2 - x_1, \quad v_2 = y_2 - y_1, \quad v_3 = z_2 - z_1$$

The standard basis vectors in $\mathbb{R}^3$ are $i = \langle 1, 0, 0 \rangle$, $j = \langle 0, 1, 0 \rangle$, and $k = \langle 0, 0, 1 \rangle$. Thus, $v = v_1 i + v_2 j + v_3 k$.

Key Properties of Vector Operations

Given vectors $u = \langle u_1, u_2, u_3 \rangle$ and $v = \langle v_1, v_2, v_3 \rangle$ and scalar $c \in \mathbb{R}$:

- Vector Addition:** $u + v = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$
- Scalar Multiplication:** $c u = \langle c u_1, c u_2, c u_3 \rangle$
- Magnitude (Euclidean Norm):** $\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$
- Unit Vector:** A vector with length 1. For any non-zero vector v , the unit vector in the direction of v is $u_v = \frac{v}{\|v\|}$.

10.2 THE DOT PRODUCT (SCALAR PRODUCT)

The dot product yields a scalar quantity and measures the extent to which two vectors point in the same direction.

Algebraic Definition

For vectors $u = \langle u_1, u_2, u_3 \rangle$ and $v = \langle v_1, v_2, v_3 \rangle$:

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3$$

Geometric Definition

Where θ is the angle between u and v ($0 \leq \theta \leq \pi$):

$$u \cdot v = \|u\| \|v\| \cos \theta$$

Applications & Projections

10.3 THE CROSS PRODUCT (VECTOR PRODUCT)

Unlike the dot product, the cross product is unique to three-dimensional space ($\mathbb{R}^3$) and produces a vector orthogonal to both input vectors.

Determinant Formulation

For vectors $u = \langle u_1, u_2, u_3 \rangle$ and $v = \langle v_1, v_2, v_3 \rangle$, the cross product is computed as:

$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2 v_3 - u_3 v_2)i - (u_1 v_3 - u_3 v_1)j + (u_1 v_2 - u_2 v_1)k$$

Properties of the Cross Product

Property	Mathematical Formulation	Description
Anti-commutativity	$u \times v = -(v \times u)$	Reversing order flips the direction vector.
Magnitude Relation	$\ u \times v\ = \ u\ \ v\ \sin\theta$	Equal to the area of the parallelogram formed by u and v .
Parallel Test	$u \times v = 0 \iff u \parallel v$	Non-zero vectors are parallel if their cross product is zero.
Scalar Triple Product	$u \cdot (v \times w)$	Computes the volume of the parallelepiped spanned by u, v, w .

10.4 LINES AND PLANES IN SPATIAL GEOMETRY

Vector equations provide an elegant mechanism for describing lines and planes in 3D space.

Equations of a Line

Passing through $P_0(x_0, y_0, z_0)$ parallel to direction vector $d = \langle a, b, c \rangle$:

- Vector Form:** $r(t) = r_0 + t d$
- Parametric Form:**
 $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$
- Symmetric Form:**
 $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$

Equations of a Plane

Passing through $P_0(x_0, y_0, z_0)$ with normal vector $n = \langle a, b, c \rangle$:

- Vector Form:** $n \cdot (r - r_0) = 0$
- Scalar Equation:**
 $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$
- Linear General Form:**
 $ax + by + cz + d = 0$

Distance Formulas in Space

Vector algebra simplifies calculating spatial distances between points, lines, and planes:

- Distance from Point S to Line through P with direction d :** $D = \frac{\| \text{vec}\{PS\} \times d \|}{\|d\|}$

10.5 QUADRIC SURFACES & COORDINATE SYSTEMS

Quadric surfaces are three-dimensional graphs of second-degree equations in three variables. Common examples include ellipsoids, paraboloids, and hyperboloids.

Alternative 3D Coordinate Systems

For problems exhibiting rotational or spherical symmetry, non-Cartesian systems simplify calculations in multivariable calculus.

Coordinate System	Variables	Conversion to Cartesian ($\mathbb{R}^3$)
Cylindrical Coordinates	(r, θ, z)	$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$
Spherical Coordinates	(ρ, θ, ϕ)	$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$

10.6 WORKED EXAMPLE & COMPREHENSIVE APPLICATIONS

Worked Problem: Intersection & Distance

Problem: Find the distance from point $S(2, 1, 3)$ to the plane defined by $3x - 2y + z = 6$.

Solution:

- Identify coefficients of normal vector $n = \langle 3, -2, 1 \rangle$ and constant $d = -6$.
- Apply distance formula:

$$D = \frac{|3(2) + (-2)(1) + 1(3) - 6|}{\sqrt{3^2 + (-2)^2 + 1^2}} = \frac{|6 - 2 + 3 - 6|}{\sqrt{9 + 4 + 1}} = \frac{1}{\sqrt{14}}$$

- Therefore, the perpendicular distance from point S to the plane is $\frac{\sqrt{14}}{14}$ units.

Summary of Core Vector Theorems

- Cauchy-Schwarz Inequality:** $|u \cdot v| \leq \|u\| \|v\|$ for any vectors $u, v \in \mathbb{R}^n$.
- Triangle Inequality:** $\|u + v\| \leq \|u\| + \|v\|$.
- Lagrange's Identity:** $\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2$.